

PROJECT AND TEAM INFORMATION

WHERE'S MY BUS

Hardware Independent Bus Tracking System

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

The problem we want to solve and why it is important.

Students at our college often do not know whether their bus has reached their stop, has it already passed, what route it will take, or does it still has free seats, or estimated time to reach university. This uncertainty causes unnecessary waiting, missed buses, and frustration especially among freshers. College authorities have zero real-time supervision over bus activity. If the university were to tackle this issue, most existing tracking solutions depend on GPS hardware, which is costly and impractical for our campus. This project solves the problem with a hardware independent system where conductors manually update the bus's current stop. With a conductor-driven update C++ backend system, we can build a zero-hardware, highly reliable Data logging network that solves transit anxiety for students and gives administrators full operational oversight. (A centralized dashboard then lets students check real-time availability, and gives the Department Head a supervisory view of the entire transportation network.)

State of the Art / Current solution (1 pt)

How the problem is usually solved today

Today students typically rely on informal methods: calling the transport office, asking friends, or simply waiting at the stop. Some institutions use commercial GPS-based apps, but these require tracking hardware in every bus, ongoing subscription costs and reliable network. Such systems usually give only a live map location and do not provide stop-specific availability like whether seats are free, whether the bus is full, or whether it has already passed. There is no local/hassle-free solution that allows students to take a look at "status per stop" of their bus. No solution that remains hardware independent by supporting manual updates by conductors, or gives supervisors a analytics dashboard. Our system fills that gap using ordinary web technology and manual location updates, also no dedicated hardware required.

Project Goals and Milestones (2 pts)

Project general goals and milestones

To Develop a hardware independent platform that helps students easily monitor college bus availability/status and also enabling efficient management and supervision of bus activities by the department. The system will enable conductors to manually update the current stop of their assigned bus. Students can select their stop and see the busses that service it, they will be color-coded for availability: Green means available with seats, Blue means available but full, Red means a bus that has already passed and Black means a bus that is currently unavailable. The system would also provide route information, next-stop details, estimated time of arrival and bus recommendations.

Initial Milestones : Complete the system requirements, roles of users, and entire system architecture. Design the database that contains buses, stops, routes, conductors, users, and trips. Implement the basic C++ classes and OOP. Implement and test data structures that include linked lists, queues, graphs, hash tables, and priority queues.

Future Milestones : Design the dashboards for students, conductors, and department heads. Manually update the bus location and its free seat availability. Integrate frontend, C++ backend, and database. Test functionality, usability, and performance of the application. Find bugs and improve algorithms of the system. Complete project documentation and final presentation.

Project Approach (3 pts)

How we plan to articulate and design a solution. Including platforms and technologies

The system will be designed in clear layers:

HTML/CSS/JavaScript frontend for students, conductors, and supervisors.

C++ backend containing the system logic.

Data-structures and OOP layer and SQLite/MySQL database for persistent data storing.

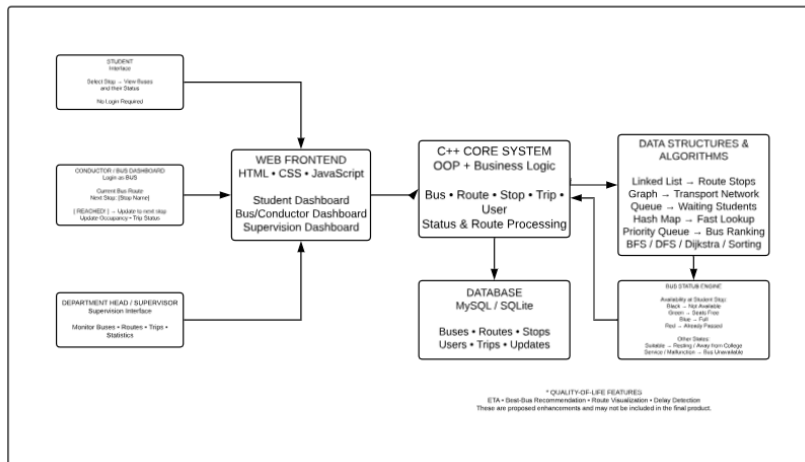
Data structures : linked lists represent ordered route stops, graphs store the model of the transportation network.

BFS/DFS and Dijkstra's shortest/fastest path : hash tables give $O(1)$ bus/route lookup.

OOP : Applied through classes like Bus, Route, Stop, and User, with encapsulation, inheritance (Student, Conductor, DepartmentHead derived from User), polymorphism, and features such as operator overloading and templates.

Technologies: C++ with STL, HTML, CSS, JavaScript, SQLite/MySQL, and Git/GitHub for version control and collaboration.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The final system is a college specific transportation dashboard where students can see which buses are assigned to their stop with color-coded availability, approximate ETA (Estimated Time Of Arrival), and a best-bus recommendation

Conductors update bus movement and status using the website and the Department Head supervises activity with trip history and analytics.

Deliverables include:

The source code (C++ logic, data structures, OOP classes, and web frontend), a populated SQLite/MySQL database, a user manual, and a final project report.

The project demonstrates practical implementation of C++, OOP, Data Structures, Algorithms, databases, and web technologies.

Assumptions

(the assumptions we are making to solve the problem)

- > *The system is hardware-independent and requires no GPS.*
- > *Location accuracy depends on the conductor updating the bus position on time.*
- > *ETA is approximate and based on stored/estimated travel times.*
- > *A network connection is needed for the web dashboard.*
- > *Real GPS, a mobile app, and push notifications are treated as future enhancements, not part of the core scope.*

References

- > *C++ and STL concepts (<https://www.w3schools.com/cpp/default.asp>)*
- > *Data Structures and Algorithms course material (TCS-302)*
- > *OOP with C++ course material (TCS-307)*
- > *MYSQL documentation and reference (<https://www.w3schools.com/MYSQL/default.asp>)*
- > *HTML/CSS/JavaScript references (<https://www.w3schools.com/html/default.asp>)*